Forbes Global 2000 (2026)

Analytics Dashboard — Project Report

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2,000	**62**	**$55.96T**	**$5.98T**
Companies Analyzed	Countries	Combined Sales	Combined Profit

1. Project Overview

This project takes the Forbes Global 2000 (2026) dataset — the world's 2,000 largest public companies, ranked by a composite of sales, profit, assets, and market value — and turns it into a complete portfolio artifact: a documented data cleaning pipeline, automated data quality tests, an exploratory analysis notebook, and a self-contained interactive BI-style dashboard.

The goal was not just to visualize the dataset, but to demonstrate a full, defensible analytics workflow end to end — the kind of process a client or employer could inspect at any stage and trust.

2. Data Cleaning Pipeline

The raw export (**data/raw/Forbes_2000_Companies_2026.csv**) contained 2,000 rows and 8 columns. The cleaning script (**scripts/clean_forbes_2000.py**) performed the following steps:

- **Split 'Headquarters' into City and Country.** One row ('Klepierre') listed only a country with no city — this was caught by an automated test and the split logic was corrected to treat single-value Headquarters entries as Country rather than silently misfiling them as City.
- **Filled 1 missing Industry value** (Medline) based on its business profile, since the field cannot be reliably inferred from other columns without domain judgment.
- **Left 1 missing Market Value as null** (Revolution Medicines) rather than imputing it — there is no reliable way to estimate market capitalization from the other financial columns, and a fabricated value would be worse than an honest gap.
- **Engineered three derived financial metrics** from source columns only: Profit Margin %, Asset Efficiency, and Return on Assets (ROA %).
- **Validated the output** against 12 automated tests covering row counts, schema, duplicate detection, value ranges, and a spot-check that recalculates Profit Margin from source columns to confirm the derived metric is computed correctly.

3. Data Quality Summary

Metric	Result
Total rows	2,000

Total columns (after cleaning)	12
Overall completeness	99.95%
Unique countries	62
Unique industries	28
Duplicate company names	0
Duplicate ranks (expected ties)	298
Automated tests passing	12 / 12

4. Key Findings

Geographic distribution

The United States accounts for 595 of the 2,000 companies (29.8%), followed by China (300), Japan (178), South Korea (66), and Canada (64). Together the top 5 countries account for over 60% of all listed companies.

Industry concentration

Banking is the single largest industry by company count (314 companies, 15.7%), reflecting how capital-intensive, regionally-fragmented banking sectors produce many nationally significant but individually smaller institutions. Diversified Financials, Construction, Insurance, and Capital Goods round out the top five.

Market value concentration

The 10 largest companies by market value — led by NVIDIA ($5.48T), Alphabet ($4.81T), and Apple ($4.41T) — represent a disproportionate share of total market value across the full 2,000-company list, illustrating how concentrated the upper tail of global market capitalization has become.

Profitability spread

Average profit margin across all companies is 20.72%, and average ROA is 5.89% — but both figures mask wide variation. A handful of companies post extreme outlier margins (e.g. Broadcom, ~2,484%) tied to one-off financial events rather than typical operating profitability, while many asset-heavy sectors (banking, insurance) post structurally lower ROA by nature of their business model.

5. Dashboard Design

The dashboard (**dashboard/index.html**) follows a BI-tool layout rather than an editorial/report layout: sidebar navigation, a KPI row, a persistent filters panel, a grid of linked charts, and a sortable company table.

Key design decisions:

- **Single visual language.** A "Trading Floor" palette — near-black background, gold accent, green/red gain-loss coding on margin and ROA figures — was chosen deliberately over a multi-theme switcher, to keep the ticker-style profitability coding meaningful and consistent.
- **Data embedded inline.** Rather than fetching the CSV at runtime, the cleaned dataset is embedded directly as JSON inside the dashboard's JavaScript. This removes a CORS/file-protocol failure mode when the dashboard is opened directly (rather than served), and means the dashboard deploys as a single self-contained unit.
- **Unified filter state.** Every filter (Country, Industry, Rank Range, Profitability tier, and search) runs through a single update function that recomputes the KPIs, every chart, the country breakdown, the leaderboard, the computed insights, and the table together — so no part of the dashboard can show a view inconsistent with the current filters.
- **Computed, not hardcoded, insights.** The Insights & Recommendations panel recalculates its narrative (market concentration, geographic exposure, margin split, and industry leaders) from whichever subset of companies is currently filtered, rather than displaying static text.

6. Testing & Validation

A pytest suite (**tests/test_data_quality.py**) validates the cleaned dataset on every run: row counts, expected columns, duplicate detection, value ranges, missing-value bounds, and a numerical spot-check that recomputes Profit Margin % from source columns to confirm the derived metric matches within rounding tolerance. This test suite caught a real bug during development — a single-value Headquarters entry that was silently misclassified during the City/Country split — which was then fixed at the source in the cleaning script rather than patched downstream.

7. Deployment

The dashboard is deployed via GitHub Pages, automated through a GitHub Actions workflow (**.github/workflows/deploy.yml**) that publishes the contents of the **dashboard/** directory on every push to the main branch. Because the dataset is embedded rather than fetched separately, the deployed dashboard has no external data dependency.

8. Skills Demonstrated

- End-to-end data pipeline design: raw data → validated, cleaned, tested output
- Root-cause debugging (fixing a data-splitting edge case at the source, not patching symptoms)

- Automated testing discipline for data quality, not just application code
- Front-end dashboard engineering: state management, dynamic chart re-rendering, computed insights
- Technical documentation: data dictionary, architecture notes, and this report
- Deployment automation via CI/CD (GitHub Actions → GitHub Pages)

Data source: Forbes Global 2000 (2026), via Kaggle. See README.md for full attribution and license notes.